KEMENTERIAN PENDIDIKAN TINGGI
JABATAN PENDIDIKAN POLITEKNIK DAN KOLEJ KOMUNITI

POLITEKNIK MALAYSIA
UNGKU OMAR

INFORMATION TECHNOLOGY & COMMUNICATION DEPARTMENT **SESSION II 2025/2026**		
DFK20013 WEB DESIGN TECHNOLOGY		
LECTURER NAME		
TYPE OF ASSESSMENT	**PROBLEM BASED ASSIGNMENT**	
TOPIC	**3.0 JavaScript Fundamentals and Interactivity with Bootstrap** **4.0 Introduction to Web and Mobile Framework**	
DURATION	**6 HOURS**	
DATE OF ASSESSMENT		
STUDENT'S INFORMATION	**NAME**	**REGISTRATION NO.**
TOTAL MARKS	**CLO 3**	**/MARKS**

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DFK20013 WEB DESIGN TECHNOLOGY			
PROBLEM BASED ASSIGNMENT			
CLO 3	Demonstrate a good practice and ethics in performing the responsive web and mobile application.	**PLO 11**	**DK DP NA (jika berkaitan)**

INSTRUCTION: This is a group assignment (3–4 students per group) where you must plan, design, and develop a mobile web application using JavaScript, Bootstrap, and jQuery Mobile, integrating interactive and mobile features with proper documentation.

SCENARIO: TECH CLUB OPEN DAY 2026

You are part of a student development team assigned to design a **mobile-friendly web application (frontend prototype)** for the upcoming **Tech Club Open Day 2026**.

The purpose of the website is to:

- promote the event
- provide clear information to visitors
- allow users to register for participation

As a developer, you are required not only to build the interface, but also to:

- apply **good practices in web design**
- consider **ethical aspects (usability, clarity, accessibility)**
- make **appropriate design decisions based on user needs**

TASKS

PART 1: PLANNING & VALUE ORGANIZATION

1. Identify:
    - Target users (e.g. students, visitors, public)
    - User needs
2. Determine at least **THREE (3) values / good practices**, such as:
    - usability
    - accessibility
    - clear information
    - user-friendly design
3. Explain:

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- How you **prioritize these values**
- Why certain design decisions are more important than others

PART 2: DEVELOPMENT (FRONTEND IMPLEMENTATION)

Develop a **TWO-page web application**:

1. **Page 1: Event Information (index.html)**

Must include:

- Header (event title + tagline)
- Event description
- **3 Bootstrap cards** (event highlights)
- Modal (for additional information)
- At least **2 tooltips**
- **Mobile integration**:
  - Call (tel:)
  - Email (mailto:)
  - SMS (sms:)
- Image gallery (3 images + external CSS styling)
- "Register Now" button (link to Page 2)

2. **Page 2: Registration Page (register.html)**

Must include:

- Registration form:
  - Full Name (required)
  - Email (valid format)
  - Phone Number (numeric only)
  - Institution
  - Category (Student / Visitor / Lecturer)
- JavaScript validation
- Success message display
- Submit button
- Back button

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PART 3: INTERACTIVITY (jQuery)

Use **jQuery** to implement:

- Document ready
- Event handling
- Page transition effect:
  - fade / slide effect between pages
- Reverse transition when navigating back

PART 4: REFLECTION & JUSTIFICATION

Explain clearly:

1. **Prioritized design vs functionality**
2. **Ethical considerations** you applied
3. Your decisions when there is conflict:
   - example: design vs usability
   - simplicity vs functionality

MINIMUM REQUIREMENTS

Your web application must include:

- 2 pages (index + register)
- Bootstrap components
- JavaScript validation
- jQuery interaction
- Mobile integration
- External CSS

EXPECTED OUTPUT

Students must submit:

- Complete web application (HTML, CSS, JS)
- Short report including:
  - planning
  - design justification
  - reflection

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PREPARED BY:	CHECKED BY:	APPROVED BY:
.....	.....	.....
(NUR HAFIZA BINTI ABD RAHMAN)	(NUR AFNI BINTI ABU HASAN)	(SAIFUL BAHARIN BIN HAIRUDDIN)
Pegawai Pendidikan Pengajian Tinggi Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar, Ipoh Perak Darul Ridzuan	Pegawai Pendidikan Pengajian Tinggi DH48 Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar	Ketua Program Diploma Teknologi Maklumat Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar
DATE : 6/4/26	DATE : 6/4/2026	DATE : 6/4/26

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RUBRIC (PLO 11, A4, CLS 5)
PROBLEM BASED ASSIGNMENT

No.	Criteria	PERFORMANCE LEVEL				Weightage (%)	Standard	Score
		Excellent 4	Good 3	Moderate 2	Poor 1			
1	**Organization of Values**	Clearly organizes and prioritizes multiple values (usability, accessibility, clarity) with strong and consistent justification throughout the project	Shows clear prioritization with some justification	Limited prioritization of values	No clear value consideration	25	/4*25	
2	**Ethical & Good Practice Application**	Consistently applies good practices (user-friendly design, clear information, validation, accessibility) throughout the system	Applies good practices with minor gaps	Basic awareness of good practices	No clear application of good practices	15	/4*15	
3	**Decision Justification**	Clearly explains and defends design decisions, especially when handling conflicts (e.g. design vs usability)	Provides some justification for decisions	Weak explanation of decisions	No justification provided	15	/4*15	
4	**Web Application Functionality**	Fully functional, responsive, and interactive (forms, navigation, components working correctly)	Mostly functional with minor issues	Basic functionality only	Not functional	15	/4* 15	

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5	**User Experience & Usability**	Highly user-friendly, well-structured, and easy to navigate	Generally user-friendly with minor issues	Some usability problems	Poor usability and confusing interface	**10**	**/4* 10**	
6	**Technical Implementation (Bootstrap, JavaScript, jQuery)**	Effective and appropriate use of Bootstrap, JavaScript, and jQuery including transition effects	Good use with minor issues	Limited use of features	No proper implementation	**10**	**/4* 10**	
7	**Reflection & Value Integration**	Deep reflection showing clear integration of values into design and development decisions	Clear reflection with some value integration	Basic reflection	Weak or no reflection	**10**	**/4* 10**	
							TOTAL SCORE	**/100**

PREPARED BY:	**CHECKED BY:**	**APPROVED BY:**
..... (NUR HAFIZAH BINTI ABD RAHMAN) Pegawai Pendidikan Pengajian Tinggi Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar, Ipoh Perak Darul Ridzuan **DATE :** 6/4/26	..... (NUR AFNI BINTI ABU HASAN) Pegawai Pendidikan Pengajian Tinggi DH48 Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar **DATE :** 6/4/2026	..... (SAIFUL BAHARIN BIN HAIRUDDIN) Ketua Program Diploma Teknologi Maklumat Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar **DATE :** 6/4/26

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